Task 1: Laying the Foundation for Analysis

Brent Oil Price Analysis: Impact of Political and Economic Events

1. Data Analysis Workflow

The analysis will follow a structured workflow to ensure the results are reliable, interpretable, and reproducible.

Step 1: Data Collection

- Load the historical Brent crude oil price dataset.
- Verify the dataset contains complete daily observations from **20 May 1987 to 30 September 2022**.
- Collect external data describing major geopolitical, economic, and policy-related events affecting oil markets.

Step 2: Data Cleaning and Preparation

- Convert the Date column into datetime format.
- Sort observations chronologically.
- Check for:
  - Missing values
  - Duplicate records
  - Incorrect date formats
  - Outliers or impossible values
- Calculate daily percentage returns for statistical analysis.

Step 3: Exploratory Data Analysis (EDA)

Perform exploratory analysis to understand the characteristics of Brent oil prices.

The analysis will include:

- Price trend over time
- Distribution of prices
- Rolling averages
- Rolling volatility
- Daily returns
- Identification of extreme price movements

Visualization methods:

- Line charts
- Histograms
- Boxplots
- Rolling mean plots
- Rolling standard deviation plots

Step 4: Time Series Analysis

Investigate important statistical properties of the series.

These include:

- Long-term trend
- Stationarity
- Seasonality
- Volatility clustering
- Structural changes

Tests to perform:

- Augmented Dickey-Fuller (ADF) Test
- KPSS Test
- Rolling Mean and Rolling Variance analysis

Step 5: Event Dataset Construction

Research major oil-market events between 1987 and 2022.

Examples include:

- Gulf Wars
- OPEC production agreements
- Financial crises
- Sanctions
- COVID-19 pandemic
- Russia–Ukraine conflict

Create a structured dataset containing:

- Event date
- Event name
- Category
- Description
- Expected market impact

Step 6: Change Point Detection

Apply Bayesian Change Point Analysis to identify dates where the statistical behavior of Brent prices changes significantly.

The model will estimate changes in:

- Mean price
- Trend
- Volatility

Detected change points will then be compared with the compiled event dataset.

Step 7: Interpretation

Compare statistically detected change points with known geopolitical and economic events.

Determine:

- Which events coincide with major price shifts
- Magnitude of changes
- Duration of effects
- Possible explanations

Step 8: Reporting

Produce:

- Statistical findings
- Visualizations
- Event timeline
- Business insights
- Investment recommendations
- Policy implications

2. Event Dataset

The following events will be compiled into a CSV file.

Date	Event	Category	Expected Impact
1990-08-02	Iraq invades Kuwait	Conflict	Sharp price increase
1991-01-17	Gulf War begins	Conflict	Increased volatility
1997-07-02	Asian Financial Crisis	Economic	Reduced oil demand

1998-03-30	OPEC Production Cuts	OPEC	Price recovery
2001-09-11	September 11 Attacks	Geopolitical	Demand uncertainty
2003-03-20	Iraq War	Conflict	Supply concerns
2008-09-15	Global Financial Crisis	Economic	Large price decline
2010-12-18	Arab Spring	Political	Supply disruptions
2014-11-27	OPEC Maintains Production	OPEC	Price collapse
2016-11-30	OPEC Production Cut Agreement	OPEC	Price increase
2018-11-05	US Sanctions on Iran	Sanctions	Supply concerns
2020-03-06	OPEC+ Price War	OPEC	Price collapse
2020-03-11	COVID-19 Declared Pandemic	Economic	Historic demand shock
2021-07-18	OPEC+ Production Increase	OPEC	Supply stabilization
2022-02-24	Russia Invades Ukraine	Conflict	Sharp price increase

Save this table as:

oil_market_events.csv

3. Assumptions

The analysis makes the following assumptions:

1. Brent crude oil prices accurately represent global crude oil market conditions.
2. Event dates are treated as the beginning of market reactions, although markets may respond before or after official announcements.
3. Each major geopolitical or economic event is assumed to have the potential to influence oil prices independently.
4. Daily closing prices sufficiently capture market responses.
5. External factors not included in the event dataset (weather, currency fluctuations, speculation, technological changes, etc.) are assumed to have smaller or random effects.

4. Limitations

Several limitations should be acknowledged.

Correlation vs Causation

The most important limitation is that statistical analysis identifies **temporal associations**, not definitive causal relationships.

A change point occurring shortly after a geopolitical event does **not** prove that the event caused the price change. Other unobserved factors may have influenced the market simultaneously.

Therefore, findings should be interpreted as evidence of **statistical correlation**, supported by economic reasoning, rather than proof of causality.

Other Limitations

- Markets often react before official announcements due to expectations and leaked information.
- Multiple events can occur simultaneously, making it difficult to isolate individual effects.
- Some geopolitical events unfold gradually rather than occurring on a single date.
- Oil prices are also influenced by exchange rates, inflation, speculation, technological advances, and macroeconomic conditions that are not explicitly modeled.
- Bayesian Change Point Analysis detects statistical shifts but does not identify the underlying cause of those shifts.

5. Understanding the Time Series

Before selecting an appropriate statistical model, several important characteristics of the Brent oil price series must be examined.

Trend Analysis

Brent oil prices exhibit long-term upward and downward trends over the study period. These trends reflect changes in global demand, production capacity, technological developments, and geopolitical conditions. Identifying trends helps determine whether the series is stationary and whether trend components should be included or removed during modeling.

Stationarity Testing

Stationarity refers to whether the statistical properties of a time series, such as its mean and variance, remain constant over time. Financial price series are typically non-stationary due to evolving market conditions. To evaluate stationarity, the following tests will be applied:

- **Augmented Dickey-Fuller (ADF) Test:** Tests the null hypothesis that the series has a unit root (is non-stationary). A significant p-value (typically < 0.05) suggests stationarity.
- **KPSS Test:** Tests the null hypothesis that the series is stationary. A significant p-value indicates non-stationarity.

Using both tests provides complementary evidence about the underlying properties of the data.

Volatility Patterns

Oil prices often experience **volatility clustering**, where periods of relatively stable prices alternate with periods of large fluctuations, especially during geopolitical crises or economic downturns. Rolling standard deviation and daily return plots will be used to examine these patterns.

Understanding volatility is important because many time series models assume constant variance. If volatility changes substantially over time, specialized models or change point detection methods become more appropriate.

How These Properties Inform Modeling

The findings from trend, stationarity, and volatility analyses guide the selection of appropriate statistical techniques. Non-stationary data may require differencing or transformation before analysis, while periods of changing variance and structural shifts motivate the use of Bayesian Change Point Analysis to detect changes in the underlying data-generating process.

6. Change Point Models

A change point model identifies moments in a time series where the underlying statistical properties change significantly. In the context of Brent oil prices, these changes may occur due to geopolitical conflicts, OPEC production decisions, international sanctions, financial crises, or major global economic events.

Rather than assuming a single consistent trend across the entire dataset, change point analysis divides the series into distinct segments, each with its own statistical characteristics, such as mean price, trend, or volatility. This allows analysts to identify structural breaks that correspond to important market events and better understand how external shocks influence oil prices over time.

7. Expected Outputs of Change Point Analysis

The Bayesian Change Point Analysis is expected to produce the following outputs:

- Estimated dates where significant structural changes occurred in the Brent oil price series.
- Posterior probabilities indicating the likelihood of each detected change point.
- Estimated parameters for each segment, such as changes in mean price, trend, or variance.
- Visualizations showing detected change points overlaid on the historical price series.
- Comparisons between statistically identified change points and known geopolitical or economic events.

These outputs help determine whether major market events align with statistically significant changes in oil price behavior.

Limitations of Change Point Analysis

While change point models are valuable for detecting structural breaks, they have several limitations:

- They identify **when** statistical changes occur but do not determine **why** they occurred.
- Closely spaced or overlapping events may make it difficult to attribute changes to a single cause.
- Results may be sensitive to model assumptions and parameter choices.
- Structural breaks caused by gradual market changes may be less clearly detected than sudden shocks.

Consequently, change point analysis should be interpreted alongside economic theory and historical context rather than as standalone evidence of causal relationships.